JQK

AI-Powered Learning Management System

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Background & Goals

⚠ Background

- **Educator Burnout:** 30% of time spent on admin tasks, grading, and repetitive emails.
- **Fragmented Systems:** Materials and Tools (AI) scattered across multiple platforms.
- **Support Gaps:** Students lack 24/7 help; professional tutors cost $30-60/hr.

🚀 Goals

- **AI Tutoring:** 24/7 intelligent support grounded in course materials (RAG).
- **Unified Platform:** Centralize dashboard for materials, tools and course management
- **Secure Architecture:** Zero-privilege data access enforcing FERPA compliance.

Intellectual Merits

- **AI mock quizzes: Test student's knowledge with unlimited quizzes generated by AI**
- **Vocabulary flashcards: Make look-up easy and instant.**
- **AI tutor: Personalized AI for each student**
- **Centralized System:** Materials and Tools (AI) concentrated in one platform.

Broader Impacts

SOCIETAL BENEFIT

Educational Equity

Democratizes access to high-quality, personalized tutoring for students who cannot afford private session

Accessibility

24/7 availability supports non-traditional students (parents, workers) who study outside standard office hours.

Privacy Standards

Demonstrates a viable model for institutions to adopt AI tools without compromising student data privacy

Microservice Architecture

Frontend

Next.js 15

Responsive UX, Role-Based Dashboards, Bilingual (EN/VI)

Backend

Spring Boot

Auth/AuthZ, Business Logic, Role Enforcement

AI Service

FastAPI + Gemini

Embeddings, RAG Pipeline, Streaming Responses

Data Layer

Mongo + Qdrant

Transactional Data & Vector Embeddings

Design Specification

Layer	Technology	Key Role
Frontend	Next.js 15.3 (TypeScript)	App Router, Shadcn/ui components, i18n support.
Backend API	Spring Boot 3.2 (Java)	Security Controller, REST API, MongoDB Integration.
AI Microservice	FastAPI (Python)	Async processing, LangChain logic, Server Sent Event Streaming.
Vector DB	Qdrant	Stores 768-dim embeddings for semantic retrieval.
LLM / Embeddings	Google Gemini 2.5	Context window management, Text-Embedding-004.

Design Specs: Security Flow

IMPLEMENTATION

Token Forwarding Implementation

The "No-Privilege" Rule:

- 1. User authenticates via NextAuth (JWT generated).
- 2. Request sent to FastAPI with JWT.
- 3. FastAPI forwards JWT to Spring Boot to fetch context.
- 4. Spring Boot validates JWT roles before returning data.
- **Result:** AI Service cannot leak data it doesn't own.

Project MileStones

I

Phase 1

Foundation

Core RAG, Auth, Microservices Setup (Complete)

II

Phase 2

Advanced Docs

LlamaParse, Recursive Chunking, PDF OCR

III

Phase 3

Automation

Automated Quiz Generation Agent

IV

Phase 4

Agents

Rubric Grading & Tutoring

We are here

Results & Finishing Tasks

STATUS

Results Achieved

Working AI Tutor Foundation: Students can ask question about their course materials with RAG implementation

Functioning Learning System: Students and Teachers have separate dashboards with login function

Smart Document Processing: System can read textbook and break into smart chunks that keep context

Finishing Tasks & Challenges

Completing AI features: Adding quiz generation features, vocabulary builder and quiz grading feature

Better Integration: Finalizing BackEnd + FrontEnd connection and AI service integration

Better Documents Reading: Adding support for images and optimizing text type document ingest

Thank you